

Sebastian Del Val Bustamante

Vehicle Dynamics & Performance Engineer

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Mechanical Engineer and incoming MSc Motorsport Engineering student at Oxford Brookes University, specializing in vehicle dynamics, performance simulation and control systems. Three years of Formula Student experience in full-vehicle modelling, telemetry, setup optimization and controls, complemented by hands-on industrial automation experience at Honda. Seeking a part-time or placement role in vehicle dynamics, performance, simulation, controls or data within motorsport.

WORK EXPERIENCE

E-Tech Formula Student Team Barcelona, ES

Head of Performance

September 2025 – June 2026

Performance Engineer

September 2023 – September 2025

- Led a **7-member Performance department**, setting technical priorities, allocating work and overseeing vehicle dynamics, simulation and data analysis projects through testing and competition deadlines.
- Developed the team's first **full-vehicle digital twin** from scratch using **IPG Carmaker and MATLAB**, enabling lap-time prediction, aerodynamic package evaluation, setup optimization and track-specific performance studies.
- Correlated simulation outputs with **real test telemetry** to validate vehicle behavior and lap-time trends and support simulation-led performance decisions.
- Modelled multi-body vehicle dynamics and extracted mass, center-of-gravity and inertia properties from SolidWorks assemblies for simulation parametrization.
- Developed and MIL-validated traction-control and torque-vectoring strategies in MATLAB/Simulink; **reduced 75 m acceleration by 0.278 s and skid pad time by 0.345 s**.
- Parametrized suspension models in MATLAB, Simulink and Simscape for a new monocoque and optimized camber/tire-pressure setup, achieving a **15% improvement** in the team's simulated lateral-grip metric.
- Developed regenerative-braking control in MATLAB/Simulink and evaluated battery SoC and energy-recovery behavior within the vehicle model.
- Analyzed **live telemetry** during endurance events at Formula Student Germany, Formula Student Spain and Formula Student Alpe Adria; also planned vehicle testing sessions and test-track layouts.

Honda Motor Company - Montesa Honda Factory

Barcelona, ES

Process Engineer

April 2025 – June 2026

- Led the end-to-end industrialization of a **Panasonic TAWERS robotic MAG welding cell**, covering technical/economic feasibility, supplier benchmarking, robot programming, fixture adaptation, quality validation, MES integration and series production.
- Brought the cell into production at **~92% OEE**; reduced cycle time by **25 s/part** on one reference and consolidated another from **two welding stages to one**.
- Validated production parts through macrographic weld inspection and 3D dimensional control, iterating welding parameters and sequence to achieve geometric and quality conformity.
- Increased daily output of a high-demand welded component from **70 to 95 units (+35.7%)** through production-data analysis and process optimisation, with similar gains on other components.
- Designed and implemented manufacturing processes for new models, adapting equipment and workflows to existing production infrastructure.
- Led automation, robotics and production-layout improvements to increase process reliability, repeatability and manufacturing flow.

E-commerce - Shopify Stores

Barcelona, ES

Founder & Operator

February 2020 – July 2024

- Founded and managed multiple Shopify stores from scratch, taking ownership of product selection, commercial strategy and day-to-day operations.
- Configured storefront UX, product pages, payment systems and digital-marketing tools across the stores.
- Used market, customer and competitor analysis to identify product opportunities and scaled an e-commerce project from scratch to **profitability**.

EDUCATION

Oxford Brookes University

Oxford, UK

MSc Motorsport Engineering

September 2026 – Present

Key Modules: Advanced Vehicle Dynamics, Laptime Simulation & Race Engineering, Composite Design & Impact Modelling, Electric Vehicles, and Engineering Business Management.

Oxford Brookes Racing (Formula Student): Vehicle Dynamics & Operations, applying vehicle dynamics, data analysis and race engineering principles to support vehicle performance and track operations.

Universitat Politècnica de Catalunya (UPC)

Barcelona, ES

Bachelor's degree in mechanical engineering

September 2022 – June 2026

Bachelor's Thesis - Grade 10/10: Development and Industrialization of a Robotic MAG Welding Cell at Montesa Honda - an end-to-end industrial project covering feasibility, supplier selection, programming, validation and production launch.

LEADERSHIP EXPERIENCE

CEU Rugby

Barcelona, ES

Rugby Captain - Forward (Second Row / Number 8)

September 2018 – September 2022

- Appointed captain for four consecutive seasons from U16 to U18, leading team cohesion, standards and motivation through competitive seasons.
- Later competed with the club's senior first team at semi-professional level in Spain's second division.

TECHNICAL SKILLS

Vehicle Dynamics & Performance: Full-vehicle modelling | Lap-time simulation | Telemetry analysis | Model correlation | Setup optimisation | Suspension modelling | Multibody dynamics | Test planning

Controls & Simulation: Traction control | Torque vectoring | Regenerative braking | MIL validation | Digital twin development

Programming & Software: MATLAB | Simulink | Python | IPG CarMaker | Simscape | SolidWorks | PTC Creo | Altair Inspire | TIA Portal | R | Excel

Languages: Spanish - Native | Catalan - Native | English - C2 Cambridge Proficiency | French - DELF B1 | Italian - Elementary